

Lecture 6 (9/25)

Variance, Normals models and Linear Regression

Variance and standard deviation

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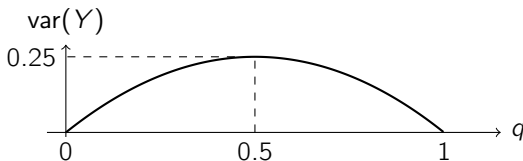
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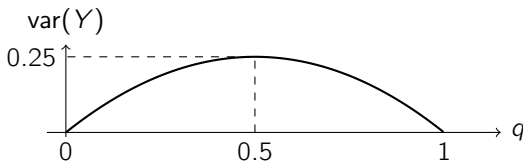
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Let $Y \sim \text{Uniform}(0, 1)$. Compute $\text{var}(Y)$.

Standard deviation

- The standard deviation is the square root of the variance,

$$\sigma = \sqrt{\text{var}(Y)} = \sqrt{E[(Y - E[Y])^2]}.$$

Unlike the variance, σ has the same units as Y , so it tells us the *typical size* of the residual $Y - E[Y]$.

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- The coefficient of variation, or CV , compares the standard deviation to the mean:

$$CV = \frac{\sigma}{\mu}$$

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Note: The CV makes sense for positive random variables. For example, it is meaningless for something like temperature.

The Normal distribution

$Y \sim \text{Normal}(\mu, \sigma^2)$ has density

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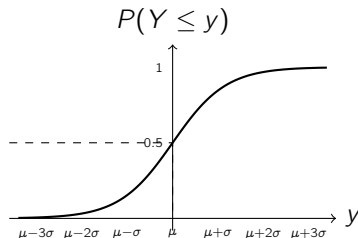
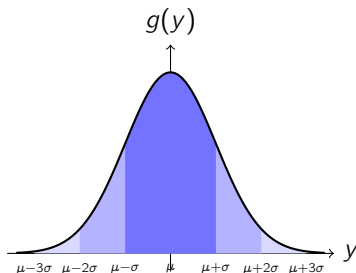
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- If X_1, X_2 are independent Normal and a, b, d are constants, then

$$aX_1 + bX_2 + d \sim \text{Normal}(a\mu_1 + b\mu_2 + d, a^2\sigma_1^2 + b^2\sigma_2^2)$$

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The linear regression model

A (single-predictor) linear regression model is conditionally Normal:

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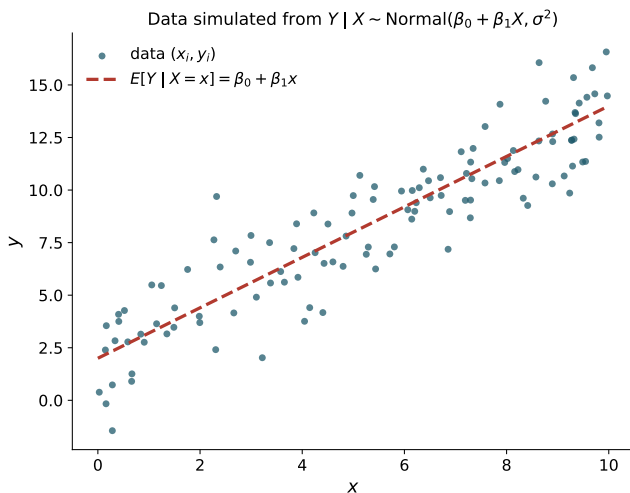
$$Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma^2).$$

Equivalently,

$$Y = \beta_0 + \beta_1 X + \epsilon, \quad \epsilon \sim \text{Normal}(0, \sigma^2).$$

What data from this model looks like

Here X has *some* distribution – the model says nothing about it, only that $Y | X$ is Normal around the line $\beta_0 + \beta_1 x$.



What the parameters mean

$$E[Y \mid X = x] = \beta_0 + \beta_1 x, \quad \text{var}(Y \mid X = x) = \sigma^2.$$

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The conditional expectation is a line in x , with slope

$$E[Y | X = x + 1] - E[Y | X = x] = \beta_1.$$

σ^2 is the spread of Y around that line, the same at every x .

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- Linearity of the conditional mean: $E[Y | X = x] = \beta_0 + \beta_1 x$.
- Exogeneity: This means $E[\epsilon | X] = 0$.
- Conditional normality: $\epsilon | X \sim \text{Normal}(0, \sigma^2)$, with the same σ^2 at every x .

Example

Let $X \sim \text{Bernoulli}(1/2)$ and $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma^2)$.

- Write $E[Y | X = 0]$ and $E[Y | X = 1]$ in terms of β_0, β_1 .
- Show $\beta_1 = E[Y | X = 1] - E[Y | X = 0]$.

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- Show $\beta_1 = E[Y | X = 1] - E[Y | X = 0]$.

For a binary predictor, β_1 is just the difference between the two group means — estimate it in data with the difference of the two group sample averages.

Example

What will the following code print?

```
g = np.random.choice([0, 1], p=[0.6, 0.4], size=2000)
s = np.empty(2000)
s[g == 0] = np.random.normal(72, 5, np.sum(g == 0))
s[g == 1] = np.random.normal(85, 5, np.sum(g == 1))

print(np.mean(s[g == 1]) - np.mean(s[g == 0]))
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Approximately 13: the difference of the two group means, i.e. $\hat{\beta}_1$.

A Normal predictor

Now let $X \sim \text{Normal}(\mu_x, \sigma_x^2)$ and $Y | X \sim \text{Normal}(\beta_1 X + \beta_0, \sigma^2)$. We can't compare group means anymore, since we'll almost never observe two samples with exactly the same x .

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$$Y \sim \text{Normal}(\beta_1 \mu_x + \beta_0, \beta_1^2 \sigma_x^2 + \sigma^2).$$

Example

For the model above, show that

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So X and Y are correlated whenever $\beta_1 \neq 0$ and $\sigma_x^2 > 0$ — this is exactly the covariance we'll use in Unit 3 to estimate β_1 from data.

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Example

A fitted model gives $\hat{\beta}_0 = 50$, $\hat{\beta}_1 = 2$. A colleague claims: “since $\hat{\beta}_1 > 0$, increasing X causes Y to increase.” Is this justified by the regression model alone? Explain in one or two sentences.